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RESEARCH ARTICLE

Integrating Fairness in the Software Design Process: An Interview Study With HCI and ML Experts

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ABSTRACT The term *Fairness* is used in the field of Machine Learning to refer to a suite of evaluation techniques, measures, and process adjustments focused on the reduction of bias in statistical models. With many different potential definitions and methodologies for implementing fairness, some contradictory in goal or incompatible in approach, there are challenges in the selection of the most appropriate definition(s) to meet the ethical expectations society or end users may have for a model. This is further confounded by the potentially complex engineering challenges in applying the selected definition. The goal of designing and developing a fair system requires interdisciplinary collaboration between user experience researchers, focused on the socio-technical challenges of application usage and adoption, and data scientists and machine learning software developers, directly involved in the technical engineering and optimisation of such models. How these two groups interact and how their areas of specialisation can support and influence the other when developing fair AI is unknown. This leaves open questions about how these stakeholders view fairness, how it can be operationalised in their current work, and what are the obstacles/opportunities for improving the treatment of fairness in practice. In order to answer these questions, 18 subject matter experts from the fields of human-computer interaction (HCI) or Machine Learning (ML) were interviewed about their experience with fairness, their perception of current fairness adoption, and the challenges they would anticipate in a set of example scenarios. This work identifies some of the current challenges to developing fair artificial intelligence (AI). In dealing with these challenges, we propose approaches that include expanding the responsibilities of HCI and ML specialists, and identifying areas where joint responsibility is required.

INDEX TERMS Fairness, healthcare, machine learning, design.

I. INTRODUCTION

The concept of fairness is one with numerous definitions. It is broadly considered the procedural ethics of decision-making [1]. It focuses, generally, on the equity and equality of groups about whom decisions are being made, as well as the process by which a decision is made about an individual. Initially the domain of the sociological and philosophical academic communities, the concept has gained prominence

within the fields of advanced statistics and machine learning (ML) from 2013 [2], with an increased volume of research from 2017 [3] as machine learning models began to see broader adoption. This emergence of fairness as a topic of discussion has been primarily driven by the challenges of creating statistical and machine-learning models that perform equally well for all demographic groups, and do not perpetuate existing systemic biases. This evaluation of fairness has been applied in cases such as a model being created that has a disproportional association of negative sentiment with advancing age, leading to classification of sentences as

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having negative sentiment due to the presence of words such as *older* [4]. Another example is the disproportional number of annotations that commercial image recognition systems (Google Cloud Vision) generate regarding a woman's physical features (e.g., smiling, hair colour) over more pertinent labels (e.g., job, demeanour) when compared to images of men [5]. Biases such as these, when encoded into ML models, have tangible negative effects on stakeholders (e.g., job applicants, patients, loan applicants), such as lower levels of accuracy for women in health care diagnosis and treatment [6].

Of the solutions proposed to deal with these challenges, the most frequently adopted by the ML community is the use of statistical equivalency metrics such as an equal rate of errors between demographic groups [7]. This includes work to enable the detection and analysis of fairness concerns [8] as well as structured processes for the analysis of fairness concerns within a data set [9]. This emerging research area of statistical analysis of fairness exists as part of a growing body of work focused on ethical computing [10], and stands alongside established work in the Human-Computer interaction (HCI) field. Initially, such HCI work focused on the accessibility of an application, ensuring its design was functional for a broad variety of users with different usage needs. More recently there has been a growing awareness of the importance of designing applications which are themselves ethical [11], via research performed in an ethical manner [12].

Designing fairer and more ethical models sits at the intersection of the statistical metrics adopted by ML subject matter experts (SMEs) and the people-centered design values upheld by the HCI SMEs. There is an emerging body of guidance on how best to integrate fairness into either the design process [13], [14] or engineering tools [14].

However, there still exists a gap in understanding how these two communities co-ordinate and support each other in the goal of developing fairer ML-enabled applications.

In this paper, we investigate how fairness is incorporated into current software development and academic research, and the priorities of experts from the human-computer interaction and machine learning communities.

To do this, we carried out a set of semi-structured interviews with 18 subject matter experts in the field of machine learning and human-computer interaction, from both academia and industry. Participants were asked about their personal perceptions of fairness, their perspectives on the current state of the art, and then to discuss in an open manner how they would incorporate fairness into the design or development of three scenarios where machine learning models were used to make decisions about humans.

Based on the analysis of these interviews, this research identifies three key themes.

Firstly, the *gathering and curation of high-quality data is a pervasive and foundational challenge*. Subject matter experts from the field of statistics and machine learning were concerned about missing data and erroneous data entries while those from the field of human-computer interaction

were concerned about biases, including racism, sexism, and homophobia, being encoded into any data that was created via historical decision making.

Secondly, *the context in which a model is used and the credence its decision is given impacts its perceived fairness*. When, how, for whom, and why a model is used deeply impacted perceived fairness more than any specific statistical or accuracy metric that focused on the models' performance.

Finally, *the metric used to evaluate the performance of the model may not be the most appropriate to evaluate the fairness of the model*. While the goal of a model may be a proximal decision such as the allocation of clinician time or the creation of treatment cohorts in healthcare support, fairness may be measured using a more distal metric such as five-year healthcare outcomes.

Building on these, we propose three areas of joint responsibility between the HCI and ML communities. The first of these three areas is user consultation and involvement in the curation and analysis of data sets as an ongoing design activity. Secondly, we propose co-responsibility in the selection and modelling of the measurement used in deciding if an ML system is performing appropriately. And finally, we expand the scope and responsibility of those advocating on behalf of the users and those engineering the models to consider the surrounding socio-technical context in which the application will be used. The themes emerging from this work reinforce the existing HCI literature regarding the need for emphasis to be placed on data curation and processing [15] and the need to adapt existing user experience (UX) tools to deal with the challenges of design for ML augmented applications [16]. To this established work, this paper contributes a clear and practical set of areas where the machine learning and human-computer interaction communities must collaborate in order to create fair and ethical ML/AI.

II. BACKGROUND

Both the ML and HCI community have attempted to address the challenges of creating inclusive ML-enabled software using the established tools and techniques of their respective domains. In this section, we will discuss the existing challenges, attempts to overcome them within the machine learning and human-computer interaction communities, as well as the intersection of these challenges between the two communities.

A. FAIRNESS AS A MACHINE LEARNING CHALLENGE

When a machine learning model is created, bias can be introduced, causing the model to be less accurate or functional for certain demographic groups [17].

These biases are introduced in numerous ways including (but not limited to) -

1) UNREPRESENTATIVE DATA SETS

Unrepresentative data sets are those that do not contain sufficient information regarding certain demographic groups. This

is done via either the unintentional or explicit exclusion of demographic considerations from data collection and analysis [18]. As an extreme example, if the training and test data is 95% composed of men, the model will likely have a high level of accuracy for that gender, but be much less accurate for other genders. Existing examples of such unrepresentative data include the omission of women from health data [19], dark skin from image libraries [20], or lower socioeconomic data from financial decisions [21]. Each of these omissions results in machine learning models with a lower level of accuracy for the identified demographic groups.

2) HISTORIC PREJUDICIAL DECISIONS

Data sets that contain a data-point or feature that is a human decision (e.g., hire or not, give medical treatment, credit worthiness) will often contain examples of historical discrimination such as racism, sexism, and homophobia necessitating not just awareness but an active anti-discriminatory approach to data analysis [22]. If the model is trained using a data set which contains discriminatory decisions the model created will, in turn, mirror this historical decision-making.

3) PROXY VARIABLES

For models that intentionally exclude demographic information to avoid it being used in the decision-making process, other variables may be a proxy for the original vulnerable variable [23]. For example, socioeconomic status can be predicted by residence location, and gender can be predicted by field of employment. These proxy variables become a defacto stand-in for demographic variables and, when used in decision-making, act as a form of hidden discrimination [24].

In the machine learning literature, fairness primarily focuses on the equivalency of errors in the generation of statistical models [7]. This has led to the creation of definitions of fairness that use metrics commonly represented in the form of a confusion matrix (Table 1) to ensure that decisions being made are equally correct between demographic groups. With the many ways and types of fairness that can be implemented, as well as the lack of an objectively correct definition to use, the selection of the most appropriate version of fairness will involve the evaluation of options, and understanding which definition will be perceived as most fair by a range of stakeholders.

There are 20+ definitions of fairness [7] that can be enforced or verified during model creation. Examples of these definitions include ensuring the likelihood of a mistake is equal between gender groups when making a medical diagnosis (Overall Accuracy Equality [25]) or that the likelihood that someone should be chosen to be on a jury should be equal across all age ranges (Statistical Parity [26]). These definitions can be impossible to satisfy simultaneously in practical environments. For example, in a hiring process, some managers may prefer to exclude demographic information from the decision-making (Fairness through Unawareness [27]) but this can be practically incompatible with ending up with a

TABLE 1. Confusion Matrix for ML.

		Predicted	
		Positive	Negative
Actual	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

representatively diverse team (Demographic Parity [1]). Due to the various definitions of what a fair model is, there is a need for explicit decisions regarding which approaches to fairness to adopt, and which to disregard.

B. WHERE HCI AND ML MEET

HCI is a field grounded in the concept of user-centered design and the inclusion of the users' perspective in the software development process. This has extended to include designing applications that are ethical [11] and performing this design work in as ethical a fashion as possible [12] by including a broad and diverse set of perspectives during the design process. Existing fairness work in the HCI field can be criticised for not attempting to fully understand what fairness means in an international context, with US-centered notions of fairness being perceived as default [28].

Domain-specific systematic reviews of machine learning implementations have called for more consideration to be given to the ethics of such applications and the use of methodologies that empower the user in the design of ML-enabled applications [29]. Established research has shown that fairness is perceived as being necessary to the creation of trust in companies or the software products those companies produce [30].

The ML communities' focus on algorithmic output alone is not enough for an application to be perceived as fair [31], with development and deployment procedures also presenting potential risks of discrimination. Research has also shown that UX practitioners can add value to the ML engineering process [16] in clarifying design goals and enabling user-centric discussions between team members. As part of this, tools have emerged from the HCI community with the goal of incorporating fairness as a fundamental part of the design of ML-enabled applications. This has included co-design checklists [13], auditing and accountability frameworks [32], and other toolkits [33] focused on the assessment and potential mitigation of perceived (un)fairness.

These processes and tools face challenges to adoption as they need to be planned for in advance and, as they have a significant and meaningful impact on the model creation process, they may require significant additional or repeated work if incorporated after the fact. Established tools such as prototyping and simulation-based iterative UX tools can be insufficient in the design of ML development [16] due to the need to understand how such ML-enabled applications will be used at an organisational level, rather than as an isolated instance. These tools also can generate functional requirements that are impractical or impossible to implement given the constraints of an ML model and data-set [16]. The

development of effective supports for designing ML-enabled applications would benefit from a better understanding of the practicalities of how engineering and design experts perceive the topic of fairness and the steps they currently take to mitigate fairness concerns (if any). There is also a gap in knowledge on what these experts require in order to do their work appropriately, and what barriers or enablers exist to the incorporation of fairness into the development process.

C. RESEARCH QUESTIONS

Based on the discussed literature, existing challenges such as poor quality and prejudicial data sets have been identified. Current attempts to overcome these challenges within the machine learning and human-computer interaction communities include approaches that focus on the levels of error in the decision-making process, or ensuring that representative stakeholders are part of the design process. While these challenges have been analysed in isolation by the respective communities, the responsibility for designing fair models is an interdisciplinary one and existing literature lacks discussion on how this joint responsibility might work.

Based on this gap that exists at the intersection of these two fields, we identified the research question -

RQ1 What challenges do human-computer interaction and machine learning experts anticipate when considering the current and future state of the art for creating Fair ML?

Answering this question also opens the opportunity to further investigate the roles and responsibilities of these two groups in dealing with these challenges. Therefore, we consider the subsequent question -

RQ2 What are the points of overlap where the ML and HCI communities are dependent on each other to meet these challenges?

III. METHODOLOGY

To answer the research questions defined for this study, we performed 18 semi-structured interviews with subject matter experts in the fields of machine learning and human-computer interaction. The study was approved by the Research Ethics Committee (REC) of the School of Computer Science and Statistics of Trinity College Dublin, reference number 20211103.

A. PARTICIPANT SELECTION CRITERIA

To qualify as a subject matter expert, the participants needed to have industry or academic experience with either HCI or ML. Academic participants needed, at a minimum, to have been or currently be active researchers in their respective field (e.g., PhD, Postdoctoral researcher, Professor); for industry participants, they needed to currently or previously have held a relevant role in software design or development (e.g., UX Designer, Data Scientist, ML Engineer).

TABLE 2. Demographic characteristics of interview participants (n=18).

Attribute	Range	Sample size
Gender	Female	10
	Male	7
	Prefer not to disclose	1
Age	25-34	13
	35 - 44	2
	45 - 54	2
	55 - 64	1
	>65	0

TABLE 3. Area of expertise among participants.

	HCI	ML	Experience with both
Industry	3	3	0
Academia	4	3	2
Experience in both	2	1	0

A screening survey was released via Qualtrics XM survey software [34]. The complete survey can be found in the Supplementary material.

B. INTERVIEW PARTICIPANTS

Among the individuals deemed eligible by the survey, 18 agreed to take part in a semi-structured interview. Demographic details for this cohort are available in Table 2. When inviting eligible participants to the interview, an effort was made to balance participants between the knowledge domains and between industry and academia. This resulted in nine HCI and seven ML subject matter experts, as well as two participants who had experience with both (e.g., UX designers for AI systems). Regarding the current field of work, nine were from industry and nine were from academia. Three participants had qualifying experience in both, having transitioned to industry from academia (e.g., had previously completed postdoctoral research roles and had since transitioned to an industry role). The participant pool leaned younger, with 13 of the 18 participants in the age range of 25-34. This may lead to participants taking approaches to design and engineering that are more tactical, as members of this age category are less likely to have senior roles in an organisation and have access to, or the remit to enforce, organisation-level systemic approaches. Further subdivisions are detailed in Table 3.

C. SEMI-STRUCTURED INTERVIEWS

In April 2022, 18 semi-structured interviews were conducted by the first author. The interviews took place over the phone or via Zoom call, depending on the participant's preference. Each interview lasted for 25-30 minutes, was audio recorded, and compensated with a €20 electronic gift card. The interviews were transcribed programmatically using transcription software [35]. These transcripts were error corrected and

anonymized by the first author. The script used for these interviews is in the Appendix.

Firstly, participants were asked about their background; those with experience in HCI were asked about their work (if any) with ML and vice versa. They were asked to share their personal definition of fairness, and if they had ever worked on projects where fairness was explicitly considered. Next, participants were shown three hypothetical scenarios where a machine-learning model was potentially being developed and adopted to replace or augment a human decision.

Quotes from these interviews are used during the analysis and discussion in this paper. These quotes are annotated with the participant number (e.g. *P12* for participant 12) and their domain of expertise and whether they come from an academic or industrial background (e.g. *HCI - Industry* for participants who work in industry within the domain of Human-Computer Interaction). Where participants discussed a specific scenario, this is included in the annotation.

D. SCENARIOS

The three scenarios were created by the research team based on identified challenges to the design and engineering of fair AI [36], [37]. The challenges were grouped into sets and a short narrative was created to frame conversations. These challenges are detailed in depth below each scenario (Sections III-D1 - III-D3). Each scenario contains distinct technical or ethical considerations that may or may not be factored in depending on the inclination and intention of the designer/developer. All three scenarios are tangentially related to the domain of healthcare. This was done to ground the discussion in a commonly understood, resource constrained, high-stakes environment. The considerations are not domain specific and apply to other areas of potential ML usage. Domain-specific analysis is potentially of value and further detailed in the section on potential future research (Sec - VI).

These scenarios were discussed with the subject matter experts in order to understand the approaches and processes they may follow in design or development. A written summary was read to each participant and a text summary was shown for the duration of the conversation. Scenarios were used to move the conversation away from general abstract fairness goals and towards more concrete, grounded conversations, to avoid the traps inherent in aspirational conversations [38]. Participants were asked to openly discuss the concerns, challenges, and priorities that they anticipated in the scenario as stated. They were then shown five definitions of fairness that were archetypal of approaches to ethical decision-making (Fairness through Unawareness [27], Fairness through Awareness [26], Overall Accuracy Equality [25], Demographic Parity [1], and Statistical Parity [26]) and asked to consider if any of these were a priority. These are detailed in Table 4, which presents the name and explanation of each definition, as well as the description given to participants. This description was the only detail shared with

TABLE 4. Definitions of Fairness shared with Participants.

Definition Name	Explanation	Described to the participant as
Fairness through Unawareness [27]	Fairness through Unawareness is a definition where any Protected Variables are intentionally excluded from the model.	Remove Demographic Factors from the decision (e.g., <i>remove gender from the data</i>)
Fairness through Awareness [26]	Fairness through awareness is a definition that uses similarities between those that the decision is being made about. It extends statistical parity to include minimising distance between subsets within demographics.	Similar people are treated similarly (e.g., <i>all men with BMI > 25 get tested</i>)
Overall Accuracy Equality [25]	Overall Accuracy Equality is an algorithmic definition that compares all base metrics. All metrics in the Confusion Matrix Table 1 are the same for all demographics.	Systems make mistakes but the mistakes should be equally distributed (e.g., <i>men and women are just as likely to be asked to do an unnecessary test</i>)
Demographic Parity [1]	Demographic Parity is a group metric that ensures the positive outcome is achieved in equal proportions by all groups.	Groups are treated differently, with the goal of achieving the same long-term result (e.g., <i>men are twice as likely to get illness X so should get twice as much funding</i>)
Statistical Parity [26]	Statistical Parity is an algorithmic definition measuring if the likelihood of a positive outcome is the same for members of socially disadvantaged demographic groups as those in the advantaged groups.	Groups are treated the same, accepting different long-term results (e.g., <i>the amount of money spent testing men should be equal to the amount testing women</i>)

the participants, they were not informed of the name of the formal definition.

1) SCENARIO ONE

The first scenario was stated as -

You are involved in the design or development of an AI/ML application that focuses on the detection of a health concern based on someone's medical history. The application will be used by doctors to proactively reach out to patients and recommend preventative care or early detection tests and will not replace routine medical engagement.

Challenging Design and Engineering Attributes - For this scenario, there are three inherent attributes that may require design or engineering changes. These attributes were not explicitly shared with the participants.

- 1) This scenario was one that was novel and proactive in nature, it is not augmenting any existing process, but instead created a new feature in an existing healthcare process. This means there are no existing benchmarks to meet before deployment.
- 2) The risk associated with false positive and false negative errors are reduced by the possibility of an incorrect decision being caught during follow-up tests, or during a routine medical examination.
- 3) As the model is supervised then there is a possibility for a "correct" or "incorrect" decision to be identified

after the fact, tracked, and incorporated back into the models' data training set for improvement.

2) SCENARIO TWO

The second scenario was stated as -

You are involved in the design or development of an AL/ML application that focuses on the hiring of doctors and healthcare professionals. The application will be used by recruiters as a part of the consideration during the hiring process.

Challenging Design and Engineering Attributes -

- 1) This scenario involves the creation of two cohorts (hired and not hired). Each with a resultant demographic makeup.
- 2) It contains no objectively correct answer with multiple potential evaluation approaches (e.g., hiring for maximum individual productivity, maximum team productivity, based on historic achievements, etc).
- 3) It is also an area where there is tension regarding whether demographics should be taken into account; there is a statistical financial performance benefit to hiring a diverse team [39] but due to some companies maintaining existing institutional goals of perceived meritocratic hiring [40] omission or obfuscation of the demographics of candidates during the hiring process is still done [41].

3) SCENARIO THREE

The third scenario was stated as -

You are involved in the design or development of an AL/ML application that decides who should receive a place in a limited capacity intensive healthcare program. The application will be used by healthcare professionals and will be the sole deciding factor.

Challenging Design and Engineering Attributes -

- 1) This is a high-stakes scenario where the decision has a significant and potentially unrecoverable effect on those for whom the decision is made about.
- 2) It involves the allocation of a finite and indivisible resource.
- 3) There is a significant variance in the damage caused to the end user between the types of errors; a false positive (receiving intensive healthcare support when it is not necessary) is not equivalent to a false negative (not receiving intensive healthcare support when it is necessary).

E. THEMATIC ANALYSIS

A thematic analysis was performed by the first author on the transcripts, to code patterns of meaning in the content

discussed following the best practices detailed by Braun and Clark [42]. A realist approach was used in the analysis of transcripts to understand the practicalities considered by a participant, especially if those practicalities were unconventional or at odds with broader best practices.

1) CODING AND THEME CREATION

Familiarisation with the data took place as the first author conducted the interviews, error-corrected the transcribed audio recordings, and read the transcripts. An initial round of inductive coding was conducted to label broad topics discussed. Then, the initial codebook was reviewed to combine duplicate and highly related codes. Following this, the second round of coding was completed to add affect and process-specific labels to enable an analysis of potential causes, thereby creating a second level of codes.

Quotes tagged with the same first-level and second-level codes were reviewed again to identify their relevance to answering the research questions and were given third-level codes as appropriate. These codes, along with the accompanying text, were then analyzed to identify foundational challenges that may have an influence on multiple codes. Drawing from the codes, the first author generated the potential themes and discussed these themes with other researchers to reflect on their applicability and appropriateness. The resultant themes are reported here.

The full codebook is available in the Supplementary material and the level 1 and 2 codes are recorded in Table 5.

F. POSITION STATEMENT

While the authors of this paper hold advanced degrees in both the fields of human-computer interaction and statistics, this research has been performed from the point of view of the HCI research community. In addition to this, the first author performing both the interviews and the primary analysis has spent time working in industry as an engineering project manager and would be perceived as being a member of socially advantaged demographic groups.

These positions may result in a framing and incidental analysis that highlights the HCI considerations and procedural engineering challenges, rather than those within current ML research or those theoretical in nature.

IV. ANALYSIS

In this section, we will report the outcomes of specific questions asked during the interviews. Following this, the themes that were generated from the coding of the transcripts are described and discussed in depth. These themes were -

- that data is a foundational challenge to fair design,
- the context and credence of a model are part of the perception of fairness,
- and that meeting a mathematical definition of fairness is not enough.

Where there were scenario-specific considerations stemming from the respective conversations (as detailed in

TABLE 5. Level 1 and 2 Codebook.

Level 1	Level 2
Accuracy	All Up Between Demographic Groups
Data Challenges	Data Drift Data Source Historical biases Missing data Missing Demographics Not everything is capturable in data Opt out of sharing Privacy Proxy variables Quality Representation in Data
Explainability	
HCI tools as a solution	
Historic Societal Issues	
Human/Machine in the loop	ML decision that contradicts human decision
Measurement modeling	
ML/AI never Appropriate	
Other FAT Concepts	
Professional Responsibility	
Remove Human Bias	
Surrounding Processes	Second Order Effects Upstream Effects
Tool Challenges	
Transparency	
Types of Errors	
What Fairness is	Difficult Choices Equal Data Equally performant No Bias No clear definition No discrimination No preferential treatment Perception Respect Unawareness Unfairness Users feel equality treated
What others are doing	Optimistic Pessimistic

Section III-D) these are highlighted. The themes were created based on the analysis of the entire transcript, however, some specific questions are reported separately as they contextualize the responses that the participants gave.

A. GENERAL QUESTION DISCUSSION

1) WHAT IS FAIRNESS?

All participants were explicitly asked ‘What do you understand by the term fairness? Both in general and as could be applied to the design of an application/model?’

Not one of the 18 participants named a formal standardized definition of fairness (such as those present in a review [7]) when asked to share their general perception of fairness. Instead, they shared a broad suite of definitions, the most common of which focused on the effects that a decision may have on the individuals and communities involved (broadly equivalent to a definition known as *Fairness through Awareness* [26]). This wide range of fairness perceptions coalesces around a smaller number of definitions when discussing specific scenarios (as detailed in Section III-D) rather than discussing fairness in general or as an abstract concept.

Of these, the most commonly discussed was preventing models from making biased decisions, as these would result in differences in treatment across individuals.

“It’s like removing bias, you know, making sure you’re treating everyone fairly [...], that kind of thing” (P6 - HCI - Industry).

Some participants focused on the transparency of a decision-making model and suggested that a model needs to be parsable by humans as a primary requirement to be considered fair.

“If it is explainable, then I’m sure that it’s fair” (P5 - ML - Industry).

Transparency is considered a broadly related topic to fairness by academic communities with some overlap in the tools used (e.g., counterfactuals used for both [43]). However, the context that this participant was discussing was concerned with the individual fairness of being able to explain and understand the decision, which is the focus of the field of explainable AI [44].

Two other common approaches to defining fairness were related to the absence of either preferential or discriminatory treatment from a decision-making process; the latter can be said to be focused on equity in decision-making and the former on equality. That this dichotomy emerged from the interviews is reflective of the broader terms in which fairness is viewed [45] — with equity, the act of treating people similarly, and equality, getting people to the same goal, being related but separate goals in almost all decision making processes.

“There should be some kind of requirements that do not favor one to another” (P10 - HCI - Academia).

“I would say fairness would be that different groups of people aren’t treated unjustly.” (P15 - ML - Academia).

A small number of participants discussed the ability to perceive unfairness more keenly than being able to define and perceive fairness. One participant commented that a model needs to get every decision correct to be considered fair but only one incorrect to be labeled unfair.

“It’s an offence, you only need one instance, to say ‘Oh, that’s been unfair’. Whereas with fairness, there are so many aspects to check.” (P5 - ML - Industry).

This is related to the concept of fairness being perceived as being the natural state. A fair system receives little notice or attention. In contrast to this, an unjust or unfair system/decision is an aberration eliciting an emotional response [46]. However, the scale of the unfairness is perceived differently depending on factors related to the one perceiving it, making using it as an objective measure difficult [47].

Lastly, there was a small group that commented on the impossibility of achieving a universal objective form of fairness.

“Yea, I suppose the concept of fairness is that things are equal. But is anything ever equal? It depends on your perspective.” (P12 - HCI - Academia & Industry).

This challenge to equality is based on the fact that different people will value the same outcome differently, or that they will value different outcomes from the same decision. As an example, a model may choose to allocate a certain type of hospital treatment to a patient. However, that person might prefer having a less effective treatment if it allowed them more time as an out-patient and may object to or resent the decision.

This challenge of multiple perspectives is also a concern as, of the numerous forms of fairness possible, several are statistically incompatible (e.g., both Demographic Parity and Statistical Parity are very unlikely to be satisfied by the same decision as one measures the likelihood of a positive outcome and the other measures the accuracy of the decision). As different people may perceive fairness differently or expect different types of fairness to be implemented, there will be situations where different groups expect mutually incompatible forms of fairness for the same decision. As an example of this, consider the situation where one set of users expect a purely meritocratic hiring process, where recruitment is completely focused on the achievements and historic performance, with a person’s demographics having no impact on the resultant decision (Counterfactual Fairness). A different set of users may want to consider the challenges that different demographics had to overcome to achieve their current level of success (FlipTest Fairness).

With these challenges in mind, the diverse understanding of what fairness is will lead to the need for the development of a collaborative understanding during the design and creation of an ML-enabled application and compromise between stakeholder goals.

2) OPTIMISTIC OR PESSIMISTIC?

All participants were read a general definition of fairness (before question 3 in the transcript, see Appendix) and explicitly asked *“Do you think other developers/designers are doing this? Why or why not?”*. Participants were, in equal measure, both optimistic and pessimistic about the long-term trajectory of designing fairer machine learning models.

Key to this optimism is a perceived change in the behavior of companies and institutions. This included both the increased accountability institutions faced and the diversity and inclusion initiatives driving institutional changes.

“I think companies have to be more cognizant of user needs, even from a self-preservation perspective, where they’re gonna get lit up on Twitter” (P12 - HCI - Academia & Industry).

“I think in the NGO¹ space that [equality, diversity, and inclusion] is more greatly integrated into how the work is done.” (P1 - HCI - Industry).

¹Non-Governmental Organization.

This trend is part of a broader set of increased ethical expectations being placed upon institutions both by governmental bodies, research funding bodies, and the consumers of software products. This is happening specifically within machine learning, with ethical and oversight boards becoming part of the organization design [48]. This is partly driven by a broader organizational level emphasis being placed on structured approaches to ethical review [49]. This should be contrasted with concerns regarding established cultures inside institutions, where there is a perceived lack of emphasis placed on ensuring fair and ethical work is being done.

“The older people in your organization, whatever it is, in a hospital or engineering firm, are not going to really care about any of that stuff. And then also in business, of course, there’s the shareholder imperative. But so there are all kinds of forces against, you know, I’d also, you know, it’s kind of seen as a fluffy tree hugger kind of thing” (P7 - ML&HCI - Academia).

This was compounded by the perceived solution-oriented mindsets valued by companies that focus on delivered projects and efficient technology, over considered and user-centric AI design.

“From my experience, also working with software developers in the industry, I think that, usually, the solution starts with the technology rather than a human consideration of human nature. And then it backfires.” (P2 - HCI - Industry).

The second part of this quote (‘And then it backfires.’) highlights an awareness in the software development community that current practices would benefit from considering not only the technology but also the needs of human users. There was also an academic equivalent to this consideration with the “Publish or Perish” [50] mindset, where work contributing toward the published paper was prioritised over investing time to ensure that fairness was considered appropriately.

“I think that is a minority that thinks about stuff. But it always comes down to what is your goal, and what is your field. I think, in Academia, you have the pressure to publish. So you don’t really care. You think that you don’t have the time.” (P4 - ML - Academia & Industry).

Finally, some participants viewed the challenges to gathering and curating high-quality data (discussed in Section IV-B1) as acting as a barrier to the development of fair and ethical models. A problem compounded by a perceived lack of training and knowledge among industry professionals on how best to handle the challenges of implementing fair AI.

“These are serious problems with data. Yeah. But to solve, to address this problem comprehensively, I think I, I also, as a technical expert, don’t have [solutions]” (P4 - ML - Academia & Industry).

3) USED IN CURRENT WORK?

Participants were asked *‘Have you previously worked on a project where fairness was an explicit consideration?’*

Of those that had previously been involved in a project that had considered a form of fairness, the most common approaches were related to data, and the gathering of a representative and unbiased data set.

“Start with firstly changing the data sources [to a less biased source and], having conscious training to avoid bias” (P16 - HCI - Academia).

Only one participant discussed a formal definition of fairness (Counterfactual Fairness [43]) and did so with the goal of furthering the model transparency and allowing a better understanding of the instance of a decision rather than using it as a tool to evaluate how fair a decision was.

“I want to surface out as much information about the model as I can because just one type of explanation is not reflective of what’s going on. [By using] a counterfactual example or a prototype example, you know that this instance is similar to that.” (P9 - ML - Academia).

Among the HCI participants, some equated fairness with accessibility. They discussed the need to ensure applications worked for as many people as possible and with the same goal of application usage.

“Maybe start thinking about accessibility issues and how many people you will exclude and why [the application is not] targeted correctly for the people that it’s supposed to be designed for.” (P10 - HCI - Academia).

On the other hand, some participants explicitly said they had not considered fairness in their work.

“There was no concept of fairness [in our projects], absolutely none. It was ‘we build what we want to build, we tell the users what we want, we give it to them’. And if they complain, we basically push back at them.” (P12 - HCI - Academia & Industry).

Among the industry machine learning SMEs interviewed, not a single participant discussed comparing model accuracy between demographic groups, one of the current preeminent fairness evaluation approaches [17]. Only one participant discussed analysing demographically disaggregated results during the creation of a descriptive research model.

“While doing my analysis, I would test to see if there were major differences in the effects based on gender mostly, I don’t think I did age or anything else.” (P13 - ML&HCI - Academia).

B. THEMES

In the previous section, we have alluded to the themes that have been generated during our analysis. We will now discuss each individually, highlighting the perspectives and challenges that underpin the general theme.

1) HIGH-QUALITY DATA IS A FOUNDATIONAL FAIRNESS CHALLENGE

“Most of the data sets have policies and ethical issues” (P11 - ML - Academia).

Ensuring the data used to create and evaluate a model is high-quality is a multifaceted, recurring challenge. Of all the issues discussed, data was frequently brought up and was identified as an underpinning challenge that influenced other areas of model development and design. This challenge is not unique to fairness. Poor quality or erroneous data sets have had a measurable negative impact on the overall accuracy and the potential usefulness of models if not mitigated [51]. The data challenge has a clear and unique effect on a model’s fairness.

Data-related issues were frequently mentioned by both ML and HCI SMEs. While both were concerned about the content of the data, machine learning experts focused on questions around data which had not been accurately recorded and reported during the data set’s creation. The HCI SMEs discussed historical examples of bias issues and intentional data gaps created due to discriminatory practices and the absence of data for members of minority groups. This incomplete history can also exist in situations where groups are reluctant to engage with systems that have traditionally been antagonistic or harmful to minority communities, for instance the Black, Indigenous, and People of Color (BIPOC) community’s distrust of the US healthcare system [52]). For both the ML and HCI SMEs, the concern of accuracy was the resultant challenge of these data gaps.

“So the incomplete medical history could be- could pose a risk to someone who might develop early cases of some diseases, but it will be missed because the history is incomplete.” (P2 - HCI - Industry - Scenario 1).

The largest concern was historical forms of discrimination including racism, sexism, homophobia, and transphobia that would be encoded in the models’ decision-making process and result in lower accuracy or negative outcomes for those groups.

“It’s unfair to have populations discriminated for no good reason, [due to] the data that was used to train [the model]. So the main issues I can see from my side would be the data” (P8 - HCI - Academia).

Also discussed was the topic of missing data. This missing data was both the possibility of explicitly missing not only demographic information, but also the context that the demographic factor may provide.

“So if you’re missing a person’s ethnicity in that dataset, and then you’re looking at the BMI as an indicator[...]. Already, you’re reading that data wrong, because it’s, you know, it’s different for different groups.” (P1 - HCI - Industry).

Lastly, errors and mistakes within the dataset were a concern. This included poorly encoded or transcribed data, or mistakes related to historically underdocumented decisions

(e.g., frequently underdiagnosed or misdiagnosed healthcare concerns)

“I’m working on [Endometriosis, it] is more commonly a disease that affects women, but also trans and nonbinary individuals. The healthcare system has ignored it and avoided it so much. And also has a long history of calling it hysteria.” (P16 - HCI - Academia).

These three data concerns are interconnected as historically disadvantaged social groups are more likely to be unrecorded or underrepresented in a data set, and also most likely to have erroneous or inaccurate decisions made about — even when unintentional — due to systemic and institutional biases. This was most clearly discussed during the analysis of the recruitment scenario (Detailed in Section III-D2). Indeed, the data sets and model are subject to historic decision-making and therefore will likely be biased to significantly favour recruitment standards that mirrors existing demographic groups within the company.

“[Women] would automatically be kind of put down a peg because they will never be seen in one of the examples resumes that they’re given to train.” (P13 - ML&HCI - Academia - Scenario 2).

2) THE CONTEXT IN WHICH A MODEL IS USED AND THE CREDENCE IT IS GIVEN IMPACTS THE PERCEPTION OF FAIRNESS

The second theme that underpinned discussions with the subject matter experts was the importance of clarity regarding the context that the model was being used and the credence it was given.

“Well, I think my first question will be at what stage of the process is this happening?” (P12 - HCI - Academia & Industry).

Regarding credence, participants were concerned about how much priority would be placed on the decision a machine learning model might come to. There were concerns regarding whether some users may become over-reliant on the decision made by the machine learning model and other users may disregard the suggestion entirely, leading to inconsistent adoption.

“[The hiring ML recommendation] can be used differently by different recruiters. And I think that’s where the concern would be. Yeah. You know, if it’s something that should be a small element of the process, I think that should be very explicit for any user of us that this should you know, this should have this weight in your total decision.” (P1 - HCI - Industry - Scenario 2).

Among the participant pool, there was a reluctance for the model to be a hard gate for potential hiring decisions. Instead, participants expected consistency and clarity about when and how it would be used. Concerns were raised regarding some users becoming over-reliant on it, effectively treating it as a

shortcut heuristic and allowing it to become an effective hard gate.

“This will probably completely depend on the person, but how much do they, again, trust [the model]. How much do they accept it? How much do they take it on board, even though it’s only one piece of information?” (P6 - HCI - Industry - Scenario 2).

Participants were, for the majority, in favour of a human-in-the-loop approach where either the machine learning model was a factor in the human decision-making process, or a human acted as a final review before a model’s decision was communicated to the intended recipient.

“Again, all of these things I would consider to be tools to support the professional but never to replace the professionals’ decision-making capacity.” (P4 - ML - Industry).

In a high-stakes context, participants strongly stated that such critical decisions (e.g., allocating a bed in intensive care, see III-D3) should not be abdicated entirely to a ML model.

“No, no, no, please, please don’t. Don’t let [the model] decide.” (P3 - HCI - Academia - Scenario 3).

The only high-stakes situation where this may be considered for some participants was when the model could demonstrate a lower level of bias than a human when making such decisions. There is also some consideration given to the fact that a model like this may remove some emotional labour from the professionals involved in critical, potentially highly demanding decisions.

This was not the only consideration given to the emotional labour that may be created or mitigated depending on the context of the model. Participants wanted the communication of a model’s decision to be conveyed by an actual person to ensure that the person receiving the message had context, instead of it seemingly coming out from nowhere and without warning.

“I don’t think that- that a system like this, at the moment, should ever go straight to a patient, [...] I would want it to be augmenting clinicians’ understanding rather than replacing a clinician.” (P5 - ML - Industry - Scenario 1).

The final approach considered for ML adoption was that of the model acting as a dissenting opinion. Such model would only notify a human when it disagreed with the opinion or the statement that they were making; in all other cases, the model would run silently and unobtrusively in the background, with no input in the decision making process.

“Maybe the tool will highlight something that the doctor didn’t consider. ‘Why not take a minute to consider that again?’ [...] It’s just a challenge before a final decision was made.” (P1 - HCI - Industry).

3) MEETING A MATHEMATICAL DEFINITION OF FAIRNESS DOES NOT MEAN THE MODEL IS ETHICAL

Building on the concepts discussed in the theme of context and credence (Section IV-B2), a fair model and a high-accuracy model were not inherently seen as being the same thing.

“It probably is fairer, whether it’s right or wrong is- is another huge discussion.” (P12 - HCI - Academia & Industry).

When discussing specific scenarios, the evaluation and fairness of the decision were talked about together. The selection of the most appropriate approach to evaluate a model was seen as a contributing factor to how fair the model was. In the scenario focused on the allocation of limited healthcare resources (III-D3), participants discussed specifics of the models’ measurements. This included concerns regarding whether the goal was optimising for minimising the total cost of treatment or expected long-term healthcare outcomes. There was also the requirement to consider weighting decisions towards specific demographic groups (e.g., prioritising children regardless of other considerations).

“If you did find that it was biased towards certain demographics or people would have to decide, do we bias it towards younger people? Or not? Or do we just look at the strict numbers to be looking at weight, and the respiratory workings, and the heart function, and things like that? Or do you take into account if somebody is disabled?” (P16 - HCI - Academia - Scenario 3).

A scenario such as this is an example of a situation where the metric used to statistically optimise the model (e.g., overall accuracy) is not the element that matters the most to users (e.g., child mortality). This highlights the need to ensure that the criteria that a model is being optimised for are the same criteria that would matter to an end user should the model be adopted.

“We want people to engage, but then, like, we have found through our own research, you know, looking at user data, that high engagement does not always mean someone’s getting better.” (P6 - HCI - Industry).

Measurement Modeling, grounded in quantitative social science research, has been proposed as a method for adding clarity to the goals used in the analysis of fairness [53]. It refers to the framework of deciding how and what to evaluate during system design. When adopted and used as part of the model development process, it provides a potential approach to weighing user preference with model optimisation.

V. DISCUSSION

In this section, we first discuss how the findings of our analysis answer the research questions. We then review the emergent areas of joint responsibility between the HCI and

ML communities, and the changes needed to enable these groups to center the user in the development of ML-enabled software.

A. RQ1

The first research question was stated as - *What challenges do Human-Computer Interaction and Machine Learning experts anticipate when considering the current and future state of the art for creating Fair ML?*

Two challenges were identified in the analysis: data-related challenges and difficulties around defining fairness.

The lack of high-quality data was a pervasive theme. This was seen as an influencing factor at all levels of the development process by the ML SMEs and its effect was an area of key concern for the HCI SMEs.

The need for clarity on what will be perceived as fair by end users was also seen as an established and as yet unresolved issue. How to appropriately identify the context in which one should use the model and which numeric measures they should use to evaluate its fairness are challenges requiring tools from both domains to answer.

B. RQ2

The second research question was stated as - *What are the points of overlap where the ML and HCI communities are dependent on each other to meet the identified challenges?*

All three of the themes generated by our analysis highlight the need for collaboration between both the design and engineering groups involved in the development of ML-enabled software.

Data ownership and management is inherently a human-centric management activity, with the surrounding context requiring constant consideration. This human-centric work should be done jointly with 1) the ML responsibility of data curation and pre-processing to address the issues of unrepresentative or biased data, and 2) the user experience responsibility of contextualising the data and ensuring that it is reflective of the system users.

With regards to the context of use of a model, we identified that the development of ML models needs to integrate within existing human workflows, rather than a model being created and the existing human workflows needing to adapt to it. In addition to this, credence underpinned the perceived fairness and appropriateness of all the machine learning models, suggesting that as models are adopted, the amount of trust their decision is given should be intentionally designed for.

Finally, our findings indicate that the fairness of a model can be evaluated at two distinct but interconnected levels. Firstly, there are the notations of fairness concerned with accurate models that are focused on the immediate decision, ignoring the long-term outcomes of such a decision. These are generally considered the responsibility of the ML specialist as they focus on direct model results, such as the rate of false positives or the overall accuracy. Secondly, there are the notations of fairness concerned with the different medium

and long-term outcomes, and the process by which these outcomes are reached. These notations are not currently standard in ML development, but user experience tools are more appropriate to understanding these socio-technical processes. These two evaluation levels will influence each other, and tradeoffs between the two will require collaboration between the SMEs.

C. AREAS OF JOINT RESPONSIBILITY

The three themes detailed all exist at the intersection between the ML responsibilities of engineering and delivering an accurate model, and the HCI responsibilities of user agency advocacy and consideration. As such, we explore the joint responsibility between the groups to address these concerns and provide a base for future fairness analysis and advocacy work.

To this end, we propose three areas as potential important directions for greater co-responsibility, including data analysis, model evaluation, and socio-technical consideration.

1) DATA GATHERING AND ANALYSIS IS A JOINT RESPONSIBILITY

Firstly, the data used in the training of ML models should be gathered and cultivated with end users in mind, and in consultation with appropriate demographic groups to fully understand any historical biases that may be included within the data set.

The importance and difficulty of this challenge suggests that a more proactive approach should be taken. This could involve the tracking and management of a curated training and evaluation data set for the duration of the models' usage using a structured form of user-centered data management [54]. This management should be done as a partnership between data scientists and user advocates to ensure that, over the lifetime of a model's usage, model drift and data drift are accounted for and do not result in additional fairness issues. This is also important in an environment where a model may be applied to a demographic group from which the data was not gathered, and a process of user-centered data management should be used to ensure this drift in usage is taken into account.

This goal stands in contrast to existing data minimisation strategies employed by teams in privacy conscious environments. Current best practices (e.g., GDPR) have included or suggested that data gathered should be kept to a minimum to protect individual privacy and minimise exposure in case of a security breach. However, without demographic information, it will not be possible to create the disaggregated results required to assess the fairness of a decision. This tension will require a reassessment of modern data gathering and storage practices.

For human-computer interaction specialists, the data collected to be used in creating a ML model should be part of the conversation when consulting end users regarding the

system. For machine learning experts, the use of a model should be reevaluated when redeployed to address the areas and demographic groups that were not part of the original assessment.

2) MODEL EVALUATION AS A HUMAN-CENTERED CHALLENGE, NOT A STATISTICAL ONE

Measurement Modeling focuses on identifying measures that can be used to evaluate and compare outcomes, as defined by the user, agnostic of the immediate decision being made. The reliability and validity of these measures are assessed during the design process [53].

The metrics used to evaluate a model are likely to differ from the measures used to assess its fairness. For instance, a model's suitability may be evaluated on its accuracy in placing participants in a treatment program that maintains their engagement the longest, while the model's fairness is assessed on 6-month healthcare outcomes. The ML literature has historically focused on the immediate decision-making process, with early analysis of fairness goals measured by comparing the immediate impact of these decisions. This focus on the accuracy of the immediate decision is understandable; it is something that an engineer can influence with changes to the model, and it is within their immediate control. However, while accuracy is seen as an important foundational metric to fairness, a highly accurate model will not necessarily be perceived as fair, as fairness is also impacted by other factors.

Assessing a model therefore comprises 1) measuring how well it achieves its goal (sometimes referred to as the evaluation criteria), and 2) determining its fairness through measurements assessing the effect the model may have on different demographic groups. This suggests the need for a split between the evaluation metrics used to minimise error-based bias, and the fairness metrics used to evaluate equity in the quality of the solution.

The choice of fairness measurement should be informed through discussion with stakeholders based on their priorities and perspectives, whereas the choice of accuracy metric will be part of the engineering development. While decoupled, these metrics will have an influence on each other, requiring shared approaches to evaluation and analysis.

3) THE SOCIAL CONTEXT SURROUNDING THE MODEL SHOULD INFORM THE MODEL'S DEVELOPMENT

The analysis highlighting the need to understand a model as part of the overall decision-making process lends further support to the argument that there should be a shift away from viewing the model in isolation from the social context in which it is used. The broader socio-technical process, including the level of credence given to the model and the context in which users engage with it, should be designed for and considered in advance of model development and optimisation. This level of consideration and credence that the model is given is not binary (e.g., the

model either being entirely in control or not in control at all), it is instead a sliding scale with levels of influence that the model may have. This includes the potential alternative of a model with no influence other than the ability to notify a decision-maker when it disagrees with a decision. This approach maintains user agency but benefits from the potentially increased accuracy stemming from a model's decision-making.

To do this, the model creation process should involve user experience researchers and designers, enabling potential users to define how and when the model should be used. This is in contrast to solutionist approaches, common within the engineering field, where models are created and leveraged without specific consideration given to the context.

VI. LIMITATIONS AND FUTURE RESEARCH

In the interest of understanding a broad set of perspectives, this qualitative research focused on gathering a small participant pool not controlled for demographic variables — other than gender identity. Therefore, participants were not asked to provide further demographic details which might have identified whether they belonged to minority/disadvantaged groups (e.g., ethnicity, sexual orientation, etc.). As these groups are best positioned to understand the complex social nuances of systems continuing to perpetuate historical biases, the omission of gathering specific feedback from these groups is both a limitation of this research and an opportunity for future work.

Additional research can focus in-depth on a specific aspect of the HCI and ML communities and how they currently support fair ML. There is also significant merit in looking into how these two communities overlap and their engagement in practice as perhaps a form of participatory action research. Another consideration for potential future work is having similar discussions with participants in senior roles. While the seniority and time in role skewed lower in our study, we can anticipate other potential approaches to fairness that are only available to those in strategic and organisational guidance roles — which would not be available to those in more tactical roles. In addition to this, a more quantitative approach to the analysis of the gaps in supporting SMEs' experience when developing fair and ethical ML-enabled applications would allow for more concrete challenge identification to better support the communities involved.

In this research previous fairness knowledge was not controlled for. As the field of fairness grows into its own academic and industrial subdomain with research and jobs in the field growing in popularity, the recruitment of a cohort who are subject matter experts in fairness may allow for a deeper and more nuanced exploration of the current state of the art. Other opportunities for future work could include domain-specific and machine learning model-specific replications of this study. Analysing the approaches SMEs take at a domain-specific level (e.g. healthcare, insurance, etc.) may generate more actionable best practices for those

fields. Similarly, research focusing on a specific type of ML technical model (e.g. Clustering, Regression, etc.) may provide insights into what is possible in a given technical approach.

Future work could also include a more formal mapping of the responsibilities of these stakeholders, including a rigorous set of guidelines and best practices, to allow these two groups to better collaborate and coordinate on the development of user-centred machine learning models. This work informs the design of future tools to support this collaboration, identifying that they will need to enable fairness analysis beyond just statistical comparison. Such tools could incorporate data curation, accuracy/fairness trade-off comparisons, and measurement modelling.

VII. CONCLUSION

This study explored the thoughts, concerns, and priorities of subject matter experts from the machine learning and human-computer interaction communities, with regard to the adoption of fair machine learning models. Unsurprisingly, fairness is subjective and its perceptions when talked about in a general context are broad and encompass a wide variety of factors. While this perception does narrow when discussing specific ML implementations (for instance in the proposed scenarios), the perceived scope is still very broad and sometimes contradictory. Regarding the academic and industrial environments in which these professionals operate, the subject matter experts are optimistic and pessimistic in equal measure. While both groups perceived the trend towards fairer machine learning as positive, the institutional and systemic barriers of change reluctance, business prioritisation, as well as a lack of training in the field tempered some of the positivity.

The key themes discussed were that 1) fairness involves a complex network of perspectives and priorities, 2) the lack of high-quality data is a challenge which underpins a broad set of other foundational concerns in the creation of ML, and 3) the context in which it is used and the credence that is given to a model impacts its perceived fairness. The goal of delivering fair and ethical AI is large and nuanced, spanning the entire software development lifecycle and beyond into the broader social context in which the system will be deployed.

To deal with the challenges identified, we discussed the importance of joint responsibility between the ML and HCI communities. Key to this is the three areas of evaluation, data gathering, and contextual adoption. Evaluation could be considered at two separate but equally important levels: the assessment of the model's fairness, and the evaluation of its immediate accuracy, both with measurements informed by the users' perspective. Data gathered can be treated as a user-centric action and proactively reviewed and managed, in consultation with the user experience experts, over the course of the model's life. Lastly, the context in which a model is used and the credence it is given should be addressed in the design process, with consideration given to the socio-technical environment it will be deployed in.

APPENDIX. SEMI-STRUCTURED INTERVIEW QUESTIONS

Interview Script

Hello. Thank you for agreeing to participate in the interview, we greatly appreciate your help with our research. My name is [...] and I will conduct this interview. The interview will last around 30 mins and will be recorded.

The recording is to accurately record the information you provide and will be used for transcription purposes only. After the transcription, we will destroy the recording. If you don't wish to be recorded you can stop the interview.

If you feel any discomfort, please, let me know and we could pause, take a break, or stop the interview.

Some guidance on the interview -

Please do not name any third parties as we will have to censor them.

Be aware that if you mention any illegal activities, I will have to be reported to the authorities.

We will be asking questions related to work you have previously done as part of your employment and/or research. In answering these questions please do not share anything that may violate non-disclosure agreements or ongoing unpublished work. Instead, feel free to discuss industry trends as you may perceive them, discuss concepts you have experience with, or you can discuss specifics but in a manner that meets any secrecy obligations that you may have.

Lastly, we will be talking about Fairness as a concept. This is something that is usually understood intuitively and can be difficult to articulate. Answers such as *"I don't know"* and just thinking out loud without a clear conclusion is fine.

Are you ready to start?

*****Start Recording*****

Introduction

1. What is your profession/field of study?

For ML SMEs-

- a. Have you worked on applications ML where the end product made decisions about people?

For Design SMEs -

- b. Have you designed applications that have a Machine Learning component?

2. What do you understand by the term fairness? Both in general and as could be applied to the design of an application/model.

Intro on ML Fairness -

We can get into more specific definitions later but here is a general definition of Fairness to start.

Fairness is generally considered to be the idea that similar people should be treated similarly. This can be defined in multiple ways and in multiple points of view. Some forms of fairness concerns can be quantified numerically such as problems of unequal error rates and unequal outcomes. Others require an examination of what should or should not be considered when making decisions.

3. Have you previously worked on a project where Fairness, as described above, was an explicit consideration? Please discuss.
4. If you were asked to incorporate Fairness in your design or model, how would you start?
5. What would you need in terms of process, tools, support, etc to be able to deliver this?
6. Do you think other developers/designers are doing this? Why or why not?

Scenarios

Introduction of fairness options -

(Screen will be shared with participants, showing the scenarios and fairness options)

Scenario One - You are involved in the design or development of an AI/ML application that focuses on the detection of a health concern based on someone's medical history. The application will be used by doctors to proactively reach out to patients and recommend preventative care or early detection tests and will not replace routine medical engagement.

What forms of fairness would you personally prioritise?

What additional information would you like?

Scenario Two - You are involved in the design or development of an AI/ML application that focuses on the hiring of doctors and healthcare professionals. The application will be used by recruiters as a part of the consideration during the hiring process.

What forms of fairness would you personally prioritise?

What additional information would you like?

Scenario Three - You are involved in the design or development of an AI/ML application that decides who should receive a place in a limited capacity intensive healthcare program. The application will be used by healthcare professionals and will be the sole deciding factor.

What forms of fairness would you personally prioritise?

What additional information would you like?

Tool and application conversations

Some projects break down the development stages at which an ML model can be created into 3

- 1 - Data Gathering and Analysis
- 2 - Modelling of Data
- 3 - Outcome Analysis

7. Which do you feel is the best time to ensure fairness is checked? All of the above and None of the above are options too.
8. At this stage in the project would you prefer tools that
 - a. Help you understand & visualise raw data
 - b. Measure based on personal evaluation metrics
 - c. Traffic light specifically identified Fairness Concerns based on an established definition

Final remarks

9. Are there any other questions you expected me to ask, or anything else you want to tell me?

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